

MediStock — Day 1 Design Review

1. ERD — table by table

users

Column	Type (inferred)	Constraints
id	BIGINT	PK
email	VARCHAR	UNIQUE, NOT NULL
password_hash	VARCHAR	NOT NULL
role	VARCHAR / ENUM	NOT NULL — pharmacist, doctor, manager, supplier

products

Column	Type (inferred)	Constraints
id	BIGINT	PK
name	VARCHAR	NOT NULL
strength	VARCHAR	NOT NULL

batches

Column	Type (inferred)	Constraints
id	BIGINT	PK
product_id	BIGINT	FK → products.id, NOT NULL
batch_no	VARCHAR	NOT NULL
expiry_date	DATE	NOT NULL
qty_on_hand	INTEGER	NOT NULL
cost	NUMERIC	NOT NULL

prescriptions

Column	Type (inferred)	Constraints
id	BIGINT	PK
patient_name	VARCHAR	NOT NULL
doctor_id	BIGINT	FK → users.id, NOT NULL
status	VARCHAR / ENUM	NOT NULL

prescription_lines

Column	Type (inferred)	Constraints
id	BIGINT	PK
prescription_id	BIGINT	FK → prescriptions.id, NOT NULL
product_id	BIGINT	FK → products.id, NOT NULL
qty	INTEGER	NOT NULL

dispenses

Column	Type (inferred)	Constraints
id	BIGINT	PK
prescription_id	BIGINT	FK → prescriptions.id, NOT NULL
pharmacist_id	BIGINT	FK → users.id, NOT NULL
created_at	TIMESTAMP	NOT NULL

stock_movements

Column	Type (inferred)	Constraints
id	BIGINT	PK
batch_id	BIGINT	FK → batches.id, NOT NULL
delta	INTEGER	NOT NULL (signed — negative for dispense, positive for delivery)
reason	VARCHAR	NOT NULL
ref	VARCHAR	Nullable — reference to the causing dispense/adjustment/delivery
at	TIMESTAMP	NOT NULL

idempotency_keys

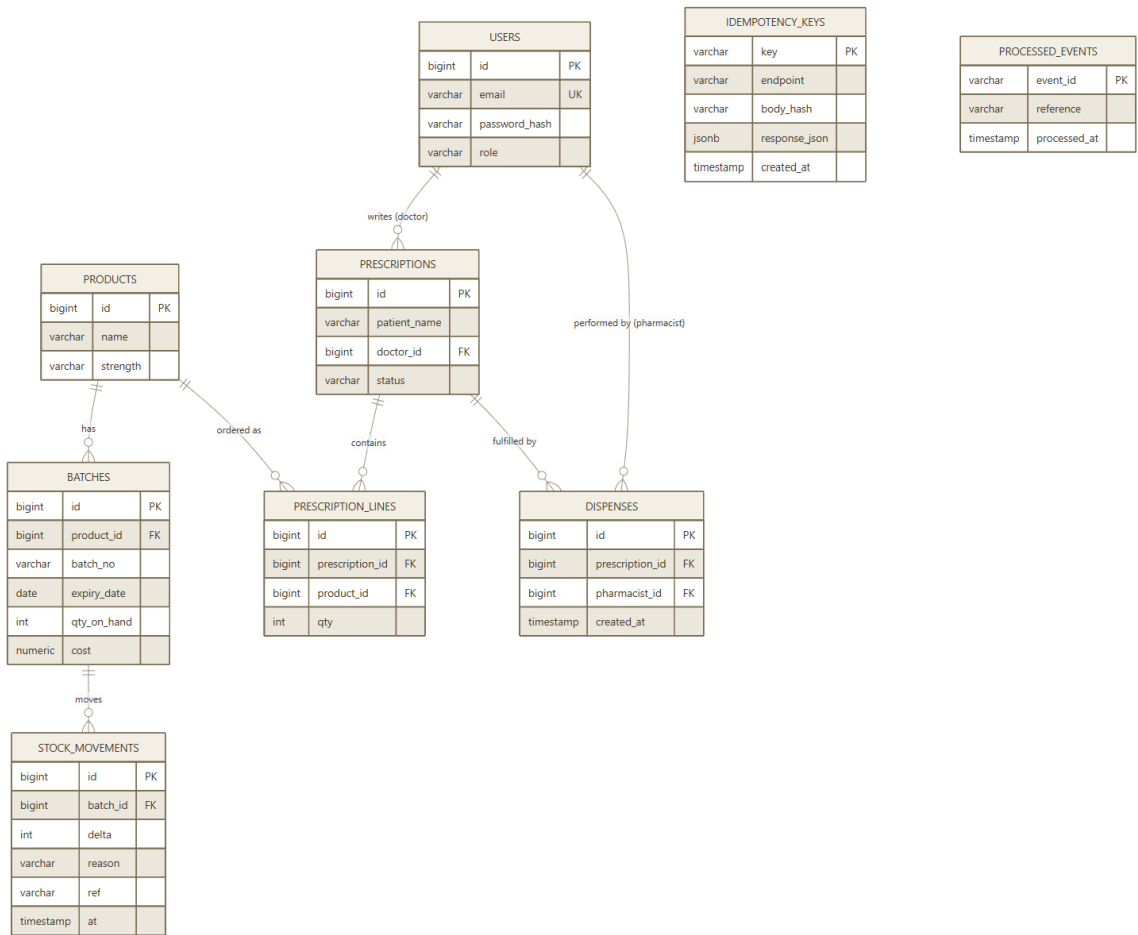
Column	Type (inferred)	Constraints
key	VARCHAR	PK — per brief
endpoint	VARCHAR	NOT NULL
body_hash	VARCHAR	NOT NULL
response_json	JSONB	NOT NULL
created_at	TIMESTAMP	NOT NULL

processed_events (for the webhook)

Column	Type (inferred)	Constraints
event_id	VARCHAR	PK (assumption — brief only says UNIQUE)

reference	VARCHAR	—
processed_at	TIMESTAMP	NOT NULL

2. Mermaid erDiagram



3. Endpoint list

Method	Path (under /api/v1)	Role	Success	Errors
POST	/auth/register	manager	201	403, 409, 422
POST	/auth/login	public	200	401
POST	/products	manager	201	403, 409, 422
POST	/products/{id}/batches	manager	201	403, 404, 422
GET	/inventory?limit&offset&low_stock	staff	200	401
POST	/prescriptions	doctor	201	403, 422
GET	/prescriptions/{id}	staff	200	404
POST	/dispenses (Idempotency-Key header)	pharmacist	201 / 200	403, 409, 422
POST	/stock/adjust	pharmacist	201	403, 404, 422
GET	/stock/movements?batch&limit&offset	manager	200	403

GET	/reports/low-stock	manager	200	403
GET	/reports/near-expiry?days=30	manager	200	403
GET	/reports/valuation	manager	200	403
POST	/supplier/deliveries (Basic auth)	machine	201	401, 422
POST	/webhooks/deliveries (signed)	machine (signed)	201 / 200	401, 422
GET	/stock/alerts/stream	manager	200 (SSE)	401, 403